

Riddhi Bandyopadhyay

Curriculum Vitae

Assistant Professor
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EDUCATION

- 2016- 2020 **Ph.D. in Physics**, University of Delaware, Newark DE, USA
Thesis supervisor: Prof. [William H. Matthaeus](#)
- 2014- 2016 **M.Sc. in Physics**, Indian Institute of Technology (IIT) Kanpur, India
- 2011- 2014 **B.Sc. with Physics Honours**, University of Calcutta, India

PROFESSIONAL EMPLOYMENT

- 2025- present Assistant Professor
Department of Physics and Astronomy, **University of Delaware**, Newark DE, USA
- 2021- present Associate Research Scholar
Department of Astrophysical Sciences, **Princeton University**, Princeton NJ, USA
- 2020- 2021 Postdoctoral Research Associate
Department of Astrophysical Sciences, **Princeton University**, Princeton NJ, USA
Group of Prof. [David J. McComas](#).
- 2020- 2020 Postdoctoral Researcher
Department of Physics and Astronomy, **University of Delaware**, Newark DE, USA
Group of Prof. [William H. Matthaeus](#).

HONORS AND AWARDS

An article titled “Collisional-like dissipation in collisionless plasmas” was chosen as a Featured Article in [Physics of Plasmas](#). Companion science highlight published in [Scilight](#).

2021 [AAPPS-DPP U30 \(under 30 years old\) Doctoral Scientist/Student Award](#) instituted by the Division of Plasma Physics, Association of Asia-Pacific Physical Societies (AAPPS-DPP).

2020 [Parvez Guzdar Young Scientist Award](#) instituted by the Institute for Plasma Research, Bhat, Gandhinagar, India.

2020 Daicar-Bata Award, Department of Physics and Astronomy, University of Delaware, for authoring one of the department’s three most impactful publications.

2020 [Donald L. Turcotte Award](#) from American Geophysical Union (AGU) for outstanding dissertation research that contributes directly to nonlinear geophysics.

2020 Qaisar and Monika Shafi Theoretical Physics Outstanding Dissertation Award, Department of Physics and Astronomy, University of Delaware.

2019 Daicar-Bata Highest GPA Award, Department of Physics and Astronomy, University of Delaware.

Innovation in Science Pursuit for Inspired Research (INSPIRE) scholarship by Ministry of Science and Technology, Government of India for a period of five years (2011 – 2016).

RESEARCH GRANT SELECTION

NASA ROSES-2020 Heliophysics Guest Investigator (H-GI) “Open” Program: “Investigation of the Nature of Turbulent Dissipation in near-Earth Space Plasma” (**PI**, 2021 – 2025).

NASA ROSES-2021 Parker Solar Probe Guest Investigator (PSP-GI) Program: “Geometry of Magnetic Fluctuations near the Sun” (**PI**, 2021 – 2025).

NASA MMS Early-Career Grant: “Ion and Electron Heating within Current Sheets in the Magnetosheath” (**PI**, 2021 – 2025).

INVITED TALKS

Pathways to Dissipation in Weakly-Collisional Plasmas, Geophysical Fluid Dynamics Session, AGU 2020.

Dissipation in Electron-only Reconnection Events: Insights from Pressure-Strain Interaction, Electron-only Reconnection Splinter Session, MMS Spring 2021 SWT Meeting.

Characterization of the sub-Alfvénic Solar Wind Observed by Parker Solar Probe, Asia Oceania Geosciences Society (AOGS) 2022 Virtual Meeting.

Energy transfer and proton-electron heating in turbulent plasmas, TESS 2022.

Energy transfer and proton-electron heating in turbulent plasmas, AAPPs-DPP 2022.

Energy Dissipation in Electron-only Reconnection, MR2023 Workshop on Magnetic Reconnection.

Turbulent Cascade and Proton–Electron Heating in Collisionless Plasmas, AGU 2023.

Studies of Multi-Scale Processes in Space Plasmas, Colloquium at the Department of Physics, University of Texas at Dallas.

REFEREED PUBLICATIONS (26 first author, [Google-Scholar](#) citations=2398, h-index=28)*

- [1] Zachary Bailey, Shadia Rifai Habbal, **Riddhi Bandyopadhyay**, and Shaheda Begum Shaik, *Quantifying Turbulence as Captured in the Inner Corona in White-light Total Solar Eclipse Images*. The Astrophysical Journal Letters, **vol. 1002**(1), (2026), p. L1. URL <http://dx.doi.org/10.3847/2041-8213/ae4d16>.
- [2] Dominic Payne, Mojtaba Akhavan-Tafti, Joshua Goodwill, Samuel Badman, *et al.*, *Evolution of Magnetic Deflections at a Conversion Layer near the Alfvén Surface*. The Astrophysical Journal Letters, **vol. 1001**(2), (2026), p. L29. URL <http://dx.doi.org/10.3847/2041-8213/ae4fbd>.
- [3] Joshua Goodwill, Subash Adhikari, Dominic Payne, **Riddhi Bandyopadhyay**, *et al.*, *Parker Solar Probe analysis across the Alfvénic transition: velocity shear, magnetic deflection, and switchback formation*. Monthly Notices of the Royal Astronomical Society, **vol. 547**(2), (2026), p. stag242. ISSN 0035-8711. URL <http://dx.doi.org/10.1093/mnras/stag242>.
- [4] Shadia Rifai Habbal, Shaheda Begum Shaik, Zachary Bailey, Nathalia Alzate, *et al.*, *Tracking the Evolution of Plasma Instabilities from the Prominence–Corona Transition Region into Interplanetary Space with Total Solar Eclipse and PSP/WISPR White Light Images*. The Astrophysical Journal, **vol. 998**(1), (2026), p. 51. URL <http://dx.doi.org/10.3847/1538-4357/ae2ec7>.
- [5] Subash Adhikari, **Riddhi Bandyopadhyay**, Joshua Goodwill, William H. Matthaeus, *et al.*, *Characterization of the Alfvén Transition in the Young Solar Wind using Parker Solar Probe Observations Approaching Solar Maximum*. The Astrophysical Journal, **vol. 997**(2), (2026), p. 259. URL <http://dx.doi.org/10.3847/1538-4357/ae2c78>.
- [6] M. B. Khan, M. A. Shay, S. Oughton, W. H. Matthaeus, *et al.*, *Factors controlling the statistics of magnetic reconnection in magnetohydrodynamic turbulence*. Phys. Rev. E, **vol. 113**, (2026), p. 015207. URL <http://dx.doi.org/10.1103/yjh7-hy2s>.

*Underlined authors indicate student mentees

- [7] S. Roy, **R. Bandyopadhyay**, S. Adhikari, Y. Yang, *et al.*, *Energy transfer and conversion in magnetic reconnection: Observation and simulation*. Physics of Plasmas, **vol. 32**(11), (2025), p. 112302. ISSN 1070-664X. URL <http://dx.doi.org/10.1063/5.0278907>.
- [8] Subash Adhikari, Carlos A. González, Yan Yang, Sean Oughton, *et al.*, *Estimation of Effective Viscosity to Quantify Collisional Behavior in Collisionless Plasma*. The Astrophysical Journal, **vol. 992**(2), (2025), p. 180. URL <http://dx.doi.org/10.3847/1538-4357/ae08b4>.
- [9] **R. Bandyopadhyay**, J. Ward, D. J. McComas, N. A. Schwadron, *et al.*, *Asymmetric Tangential Velocity inside Switchbacks: Implication for Switchback Origin*. The Astrophysical Journal, **vol. 991**(1), (2025), p. 41. URL <http://dx.doi.org/10.3847/1538-4357/adfc45>.
- [10] M. E. Cuesta, G. Livadiotis, D. J. McComas, L. Y. Khoo, *et al.*, *Transfer of Entropy between the Magnetic Field and Solar Energetic Particles during an Interplanetary Coronal Mass Ejection*. The Astrophysical Journal Letters, **vol. 984**(2), (2025), p. L50. URL <http://dx.doi.org/10.3847/2041-8213/adcbff>.
- [11] H. A. Farooki, M. E. Cuesta, **R. Bandyopadhyay**, G. Livadiotis, *et al.*, *Estimating the Bulk Velocity of Energetic Particle Populations Using ISOIS/EPI-Lo Measurements*. The Astrophysical Journal, **vol. 984**(1), (2025), p. 70. URL <http://dx.doi.org/10.3847/1538-4357/adc12b>.
- [12] M. Baraka, O. Le Contel, P. Canu, S. W. Alqeeq, *et al.*, *MMS Analysis of a Dayside Compressed Magnetospheric Separatrix in the Presence of Cold Ions and a Moderate Guide Field*. Journal of Geophysical Research: Space Physics, **vol. 130**(4), (2025), p. e2024JA033234. URL <http://dx.doi.org/10.1029/2024JA033234>.
- [13] M. E. Cuesta, L. Y. Khoo, G. Livadiotis, M. M. Shen, *et al.*, *Comparing Methods for Calculating Solar Energetic Particle Intensities: Rebinning versus Spectral Binning*. The Astrophysical Journal, **vol. 980**(2), (2025), p. 235. URL <http://dx.doi.org/10.3847/1538-4357/adaea9>.
- [14] **Riddhi Bandyopadhyay**, James R. Beattie, and Amitava Bhattacharjee, *Density Fluctuation–Mach Number Scaling in Compressible, High Plasma Beta Turbulence: In Situ Space Observations and High-Reynolds Number Simulations*. The Astrophysical Journal Letters, **vol. 982**(2), (2025), p. L45. URL <http://dx.doi.org/10.3847/2041-8213/adbe3b>.
- [15] Zachary Bailey, **Riddhi Bandyopadhyay**, Shadia Habbal, and Miloslav Druckmüller, *Measurement of Turbulence Injection Scale Down to the Chromosphere*. The Astrophysical Journal Letters, **vol. 980**(2), (2025), p. L20. URL <http://dx.doi.org/10.3847/2041-8213/ada363>.
- [16] J. E. Stawarz, P. A. Muñoz, N. Bessho, R. Bandyopadhyay, *et al.*, *The Interplay Between Collisionless Magnetic Reconnection and Turbulence*. Space Science Reviews, **vol. 220**(8), (2024), p. 90. URL <http://dx.doi.org/10.1007/s11214-024-01124-8>.
- [17] David Ruffolo, Panisara Thepthong, Peera Pongkitiwanchakul, Sohom Roy, *et al.*, *Observed Fluctuation Enhancement and Departure from WKB Theory in Sub-Alfvénic Solar Wind*. The Astrophysical Journal Letters, **vol. 977**(1), (2024), p. L19. URL <http://dx.doi.org/10.3847/2041-8213/ad9727>.
- [18] O. W. Roberts, K. G. Klein, Z. Vörös, R. Nakamura, *et al.*, *Measurement of the Taylor Microscale and the Effective Magnetic Reynolds Number in the Solar Wind With Cluster*. Journal of Geophysical Research: Space Physics, **vol. 129**(11), (2024), p. e2024JA032968. URL <http://dx.doi.org/https://doi.org/10.1029/2024JA032968>.
- [19] **R. Bandyopadhyay**, N. V. Sarlis, J. M. Weygand, R. J. Strangeway, *et al.*, *Observation of chaotic fluctuations in turbulent plasma*. Physics of Plasmas, **vol. 31**(10), (2024), p. 100702. ISSN 1070-664X. URL <http://dx.doi.org/10.1063/5.0220376>.
- [20] M. E. Cuesta, A. T. Cummings, G. Livadiotis, D. J. McComas, *et al.*, *Observations of Kappa Distributions in Solar Energetic Protons and Derived Thermodynamic Properties*. The Astrophysical Journal, **vol. 973**(2), (2024), p. 76. URL <http://dx.doi.org/10.3847/1538-4357/ad68fd>.
- [21] G. Livadiotis, A. T. Cummings, M. E. Cuesta, **R. Bandyopadhyay**, *et al.*, *Kappa-tail Technique: Modeling and Application to Solar Energetic Particles Observed by Parker Solar Probe*. The Astrophysical Journal, **vol. 973**(1), (2024), p. 6. URL <http://dx.doi.org/10.3847/1538-4357/ad5e72>.

- [22] H. Hasegawa, M. R. Argall, N. Aunai, R. Bandyopadhyay, *et al.*, *Advanced methods for analyzing in-situ observations of magnetic reconnection*. Space Science Reviews, **vol. 220**(1572-9672). URL <http://dx.doi.org/10.1007/s11214-024-01095-w>.
- [23] J. T. Lang, R. D. Strauss, N. E. Engelbrecht, J. P. van den Berg, *et al.*, *A Detailed Survey of the Parallel Mean Free Path of Solar Energetic Particle Protons and Electrons*. The Astrophysical Journal, **vol. 971**(1), (2024), p. 105. URL <http://dx.doi.org/10.3847/1538-4357/ad55c3>.
- [24] Manuel Enrique Cuesta, D. J. McComas, L. Y. Khoo, **R. Bandyopadhyay**, *et al.*, *Correlation of Coronal Mass Ejection Shock Temperature with Solar Energetic Particle Intensity*. The Astrophysical Journal, **vol. 964**(2), (2024), p. 114. URL <http://dx.doi.org/10.3847/1538-4357/ad245d>.
- [25] S. Roy, **R. Bandyopadhyay**, W. H. Matthaeus, and P. S. Pyakurel, *Energy Dissipation in Electron-only Reconnection*. The Astrophysical Journal, **vol. 964**(1), (2024), p. 44. URL <http://dx.doi.org/10.3847/1538-4357/ad2769>.
- [26] C. A. González, J. L. Verniero, **R. Bandyopadhyay**, and A. Tenerani, *Local Proton Heating at Magnetic Discontinuities in Alfvénic and Non-Alfvénic Solar Wind*. The Astrophysical Journal, **vol. 963**(2), (2024), p. 148. URL <http://dx.doi.org/10.3847/1538-4357/ad1be5>.
- [27] Yan Yang, William H. Matthaeus, Sean Oughton, **Riddhi Bandyopadhyay**, *et al.*, *Effective viscosity, resistivity, and Reynolds number in weakly collisional plasma turbulence*. Monthly Notices of the Royal Astronomical Society, **vol. 528**(4), (2024), pp. 6119. ISSN 0035-8711. URL <http://dx.doi.org/10.1093/mnras/stae355>.
- [28] Panisara Thepthong, Peera Pongkitiwanichakul, David Ruffolo, Rungployphan Kieokaew, *et al.*, *Scale and Time Dependence of Alfvénicity in the Solar Wind as Observed by the Parker Solar Probe*. The Astrophysical Journal, **vol. 962**(1), (2024), p. 37. URL <http://dx.doi.org/10.3847/1538-4357/ad1592>.
- [29] **R. Bandyopadhyay**, C. M. Meyer, C. M. Meyer, Bailey C. M. Meyer, W. H. Matthaeus, D. J. McComas, *et al.*, *Estimates of Proton and Electron Heating Rates Extended to the Near-Sun Environment*. The Astrophysical Journal Letters, **vol. 955**(2), (2023), p. L28. URL <http://dx.doi.org/10.3847/2041-8213/acf85e>.
- [30] O. W. Roberts, Z. Voros, K. Torkar, J. Stawarz, *et al.*, *Estimation of the Error in the Calculation of the Pressure-Strain Term: Application in the Terrestrial Magnetosphere*. Journal of Geophysical Research: Space Physics, **vol. 128**(8), (2023), p. e2023JA031565. URL <http://dx.doi.org/https://doi.org/10.1029/2023JA031565>.
- [31] **Riddhi Bandyopadhyay**, Yan Yang, William H. Matthaeus, Tulasi N. Parashar, *et al.*, *Collisional-like dissipation in collisionless plasmas*. Physics of Plasmas, **vol. 30**(8), (2023), p. 080702. ISSN 1070-664X. URL <http://dx.doi.org/10.1063/5.0146986>.
- [32] F. Pecora, Y. Yang, A. Chasapis, S. Servidio, *et al.*, *Relaxation of the turbulent magnetosheath*. Monthly Notices of the Royal Astronomical Society, **vol. 525**(1), (2023), pp. 67. ISSN 0035-8711. <https://academic.oup.com/mnras/article-pdf/525/1/67/51088731/stad2232.pdf>, URL <http://dx.doi.org/10.1093/mnras/stad2232>.
- [33] J. L. Burch, K. J. Genestreti, S. V. Heuer, A. Chasapis, *et al.*, *Electron energy dissipation in a magnetotail reconnection region*. Physics of Plasmas, **vol. 30**(8), (2023), p. 082903. ISSN 1070-664X. URL <http://dx.doi.org/10.1063/5.0153628>.
- [34] Maruca, Bennett A., Qudsi, Ramiz A., Alterman, B. L., Walsh, Brian M., *et al.*, *The Trans-Heliospheric Survey - Radial trends in plasma parameters across the heliosphere*. Astronomy & Astrophysics, **vol. 675**, (2023), p. A196. URL <http://dx.doi.org/10.1051/0004-6361/202345951>.
- [35] N. Andrés, **R. Bandyopadhyay**, D. J. McComas, J. R. Szalay, *et al.*, *Observation of Turbulent Magnetohydrodynamic Cascade in the Jovian Magnetosheath*. The Astrophysical Journal, **vol. 945**(1), (2023), p. 8. URL <http://dx.doi.org/10.3847/1538-4357/acb7e0>.
- [36] Yan Yang, Francesco Pecora, William H. Matthaeus, Sohom Roy, *et al.*, *Quantifying the Agyrotropy of Proton and Electron Heating in Turbulent Plasmas*. The Astrophysical Journal, **vol. 944**(2), (2023), p. 148. URL <http://dx.doi.org/10.3847/1538-4357/acb25a>.
- [37] N. E. Raouafi, L. Matteini, J. Squire, S. T. Badman, *et al.*, *Parker Solar Probe: Four Years of Discoveries at Solar Cycle Minimum*. Space Science Reviews, **vol. 219**(1572-9672), (2023), p. 8. URL <http://dx.doi.org/10.1007/s11214-023-00952-4>.

- [38] S. Roy, **R. Bandyopadhyay**, Y. Yang, T. N. Parashar, *et al.*, *Turbulent Energy Transfer and Proton–Electron Heating in Collisionless Plasmas*. The Astrophysical Journal, **vol. 941**(2), (2022), p. 137. URL <http://dx.doi.org/10.3847/1538-4357/aca479>.
- [39] C. Phillips, **R. Bandyopadhyay**, D. J. McComas, and S. D. Bale, *Association of intermittency with electron heating in the near-Sun solar wind*. Monthly Notices of the Royal Astronomical Society: Letters. ISSN 1745-3925. Slac143, URL <http://dx.doi.org/10.1093/mnrasl/slac143>.
- [40] **Riddhi Bandyopadhyay**, Ramiz A. Qudsi, S. Peter Gary, William H. Matthaeus, *et al.*, *Interplay of turbulence and proton-microinstability growth in space plasmas*. Physics of Plasmas, **vol. 29**(10), (2022), p. 102107. URL <http://dx.doi.org/10.1063/5.0098625>.
- [41] Yanwen Wang, Rohit Chhiber, Subash Adhikari, Yan Yang, *et al.*, *Strategies for Determining the Cascade Rate in MHD Turbulence: Isotropy, Anisotropy, and Spacecraft Sampling*. The Astrophysical Journal, **vol. 937**(2), (2022), p. 76. URL <http://dx.doi.org/10.3847/1538-4357/ac8f90>.
- [42] J. Zhang, S. Y. Huang, Z. G. Yuan, K. Jiang, *et al.*, *Higher-order Turbulence Statistics in the Sub-Alfvénic Solar Wind Observed by Parker Solar Probe*. The Astrophysical Journal, **vol. 937**(2), (2022), p. 70. URL <http://dx.doi.org/10.3847/1538-4357/ac8c34>.
- [43] **R. Bandyopadhyay**, L. J. Begley, B. A. Maruca, D. J. McComas, *et al.*, *Beta-Dependent Constraints on Ion Temperature Anisotropy in Jupiter’s Magnetosheath*. Geophysical Research Letters, **vol. 49**(15), (2022), p. e2022GL098053. URL <http://dx.doi.org/https://doi.org/10.1029/2022GL098053>.
- [44] C. Phillips, **R. Bandyopadhyay**, and D. J. McComas, *Taylor microscale and effective Reynolds number near the Sun from PSP*. The Astrophysical Journal, **vol. 933**(1), (2022), p. 33. URL <http://dx.doi.org/10.3847/1538-4357/ac713f>.
- [45] Nikos Sioulas, Zesen Huang, Marco Velli, Rohit Chhiber, *et al.*, *Magnetic Field Intermittency in the Solar Wind: Parker Solar Probe and SolO Observations Ranging from the Alfvén Region up to 1 AU*. The Astrophysical Journal, **vol. 934**(2), (2022), p. 143. URL <http://dx.doi.org/10.3847/1538-4357/ac7aa2>.
- [46] Manuel Enrique Cuesta, Rohit Chhiber, Sohom Roy, Joshua Goodwill, *et al.*, *Isotropization and Evolution of Energy-containing Eddies in Solar Wind Turbulence: Parker Solar Probe, Helios 1, ACE, WIND, and Voyager 1*. The Astrophysical Journal Letters, **vol. 932**(1), (2022), p. L11. URL <http://dx.doi.org/10.3847/2041-8213/ac73fd>.
- [47] Sakshee Sakshee, **Riddhi Bandyopadhyay**, and Supratik Banerjee, *MHD-scale anisotropy in solar wind turbulence near the Sun using Parker solar probe data*. Monthly Notices of the Royal Astronomical Society, **vol. 514**(1), (2022), pp. 1282. ISSN 0035-8711. URL <http://dx.doi.org/10.1093/mnras/stac1449>.
- [48] Rohit Chhiber, William H Matthaeus, Arcadi V Usmanov, **Riddhi Bandyopadhyay**, *et al.*, *An extended and fragmented Alfvén zone in the Young Solar Wind*. Monthly Notices of the Royal Astronomical Society, **vol. 513**(1), (2022), pp. 159. ISSN 0035-8711. URL <http://dx.doi.org/10.1093/mnras/stac779>.
- [49] Yan Yang, William H. Matthaeus, Sohom Roy, *et al.*, *Pressure-Strain Interaction as the Energy Dissipation Estimate in Collisionless Plasma*. The Astrophysical Journal, **vol. 929**(2), (2022), p. 142. URL <http://dx.doi.org/10.3847/1538-4357/ac5d3e>.
- [50] Nikos Sioulas, Marco Velli, Rohit Chhiber, Loukas Vlahos, *et al.*, *Statistical Analysis of Intermittency and its Association with Proton Heating in the Near-Sun Environment*. The Astrophysical Journal, **vol. 927**(2), (2022), p. 140. URL <http://dx.doi.org/10.3847/1538-4357/ac4fc1>.
- [51] Francesco Pecora, William H. Matthaeus, Leonardo Primavera, Antonella Greco, *et al.*, *Magnetic Switchback Occurrence Rates in the Inner Heliosphere: Parker Solar Probe and 1 au*. The Astrophysical Journal Letters, **vol. 929**(1), (2022), p. L10. URL <http://dx.doi.org/10.3847/2041-8213/ac62d4>.
- [52] **R. Bandyopadhyay**, W. H. Matthaeus, D. J. McComas, R. Chhiber, *et al.*, *Sub-Alfvénic Solar Wind Observed by the Parker Solar Probe: Characterization of Turbulence, Anisotropy, Intermittency, and Switchback*. The Astrophysical Journal Letters, **vol. 926**(1), (2022), p. L1. URL <http://dx.doi.org/10.3847/2041-8213/ac4a5c>.

- [53] **R. Bandyopadhyay** and D. J. McComas, *Geometry of Magnetic Fluctuations near the Sun from the Parker Solar Probe*. The Astrophysical Journal, **vol. 923**(2), (2021), p. 193. URL <http://dx.doi.org/10.3847/1538-4357/ac3486>.
- [54] **R. Bandyopadhyay**, A. Chasapis, W. H. Matthaeus, T. N. Parashar, *et al.*, *Energy dissipation in turbulent reconnection*. Physics of Plasmas, **vol. 28**(11), (2021), p. 112305. URL <http://dx.doi.org/10.1063/5.0071015>.
- [55] Carlos S. Hernández, Luca Sorriso-Valvo, **Riddhi Bandyopadhyay**, Alexandros Chasapis, *et al.*, *Impact of Switchbacks on Turbulent Cascade and Energy Transfer Rate in the Inner Heliosphere*. The Astrophysical Journal Letters, **vol. 922**(1), (2021), p. L11. URL <http://dx.doi.org/10.3847/2041-8213/ac36d1>.
- [56] **R. Bandyopadhyay**, D. J. McComas, J. R. Szalay, F. Allegrini, *et al.*, *Observation of Kolmogorov Turbulence in the Jovian Magnetosheath From JADE Data*. Geophysical Research Letters, **vol. 48**(15), (2021), p. e2021GL095006. URL <http://dx.doi.org/10.1029/2021GL095006>.
- [57] E. J. Zirnstein, R. Kumar, **R. Bandyopadhyay**, M. A. Dayeh, *et al.*, *Turbulent Acceleration of Interstellar Pickup Ions at the Heliospheric Termination Shock Forms the Global ENA Spectrum*. The Astrophysical Journal Letters, **vol. 916**(2), (2021), p. L21. URL <http://dx.doi.org/10.3847/2041-8213/ac12cc>.
- [58] Bennett A. Maruca, Jeffersson A. Agudelo Rueda, **Riddhi Bandyopadhyay**, Federica B. Bianco, *et al.*, *MagneToRE: Mapping the 3-D Magnetic Structure of the Solar Wind Using a Large Constellation of Nanosatellites*. Frontiers in Astronomy and Space Sciences, **vol. 8**, (2021), p. 108. ISSN 2296-987X. URL <http://dx.doi.org/10.3389/fspas.2021.665885>.
- [59] Yanwen Wang, **Riddhi Bandyopadhyay**, Rohit Chhiber, William H. Matthaeus, *et al.*, *Statistical Survey of Collisionless Dissipation in the Terrestrial Magnetosheath*. Journal of Geophysical Research: Space Physics, **vol. 126**(6), (2021), p. e2020JA029000. URL <http://dx.doi.org/https://doi.org/10.1029/2020JA029000>.
- [60] **R. Bandyopadhyay**, W. H. Matthaeus, D. J. McComas, C. J. Joyce, *et al.*, *Energetic Particle Behavior in near-Sun Magnetic Field Switchbacks from PSP*. Astroomy & Astrophysics, **vol. 650**, (2021), p. L4. URL <http://dx.doi.org/10.1051/0004-6361/202039800>.
- [61] T. A. Bowen, S. D. Bale, **R. Bandyopadhyay**, J. W. Bonnell, *et al.*, *Kinetic-Scale Turbulence in the Venusian Magnetosheath*. Geophysical Research Letters, **vol. 48**(2), (2021), p. e2020GL090783. URL <http://dx.doi.org/https://doi.org/10.1029/2020GL090783>.
- [62] Alexandros Chasapis, William H. Matthaeus, **R. Bandyopadhyay**, Rohit Chhiber, *et al.*, *Scaling and Anisotropy of Solar Wind Turbulence at Kinetic Scales during the MMS Turbulence Campaign*. The Astrophysical Journal, **vol. 903**(2), (2020), p. 127. URL <http://dx.doi.org/10.3847/1538-4357/abb948>.
- [63] **R. Bandyopadhyay**, Alexandros Chasapis, D. J. Gershman, B. L. Giles, *et al.*, *Observation of an inertial-range energy cascade within a reconnection jet in the Earth's magnetotail*. Monthly Notices of the Royal Astronomical Society: Letters, **vol. 500**(1), (2020), pp. L6. ISSN 1745-3925. URL <http://dx.doi.org/10.1093/mnrasl/slaa171>.
- [64] S. Peter Gary, **R. Bandyopadhyay**, Ramiz A. Qudsi, William H. Matthaeus, *et al.*, *Particle-in-cell Simulations of Decaying Plasma Turbulence: Linear Instabilities versus Nonlinear Processes in 3D and 2.5D Approximations*. The Astrophysical Journal, **vol. 901**(2), (2020), p. 160. URL <http://dx.doi.org/10.3847/1538-4357/abb2ac>.
- [65] D. Ruffolo, W. H. Matthaeus, R. Chhiber, A. V. Usmanov, *et al.*, *Shear-driven Transition to Isotropically Turbulent Solar Wind Outside the Alfvén Critical Zone*. The Astrophysical Journal, **vol. 902**(2), (2020), p. 94. URL <http://dx.doi.org/10.3847/1538-4357/abb594>.
- [66] **R. Bandyopadhyay**, William H. Matthaeus, Alexandros Chasapis, Christopher T. Russell, *et al.*, *Direct Measurement of the Solar-wind Taylor Microscale Using MMS Turbulence Campaign Data*. The Astrophysical Journal, **vol. 899**(1), (2020), p. 63. URL <http://dx.doi.org/10.3847/1538-4357/ab9ebe>.
- [67] **Riddhi Bandyopadhyay**, William H. Matthaeus, Tulasi N. Parashar, Yan Yang, *et al.*, *Statistics of Kinetic Dissipation in the Earth's Magnetosheath: MMS Observations*. Physical Review Letters, **vol. 124**, (2020), p. 255101. URL <http://dx.doi.org/10.1103/PhysRevLett.124.255101>.

- [68] **Riddhi Bandyopadhyay**, Luca Sorriso-Valvo, Alexandros Chasapis, Petr Hellinger, *et al.*, *In Situ Observation of Hall Magnetohydrodynamic Cascade in Space Plasma*. Physical Review Letters, **vol. 124**, (2020), p. 225101. URL <http://dx.doi.org/10.1103/PhysRevLett.124.225101>.
- [69] Ramiz A. Qudsi, **Riddhi Bandyopadhyay**, Bennett A. Maruca, Tulasi N. Parashar, *et al.*, *Intermittency and Ion Temperature–Anisotropy Instabilities: Simulation and Magnetosheath Observation*. The Astrophysical Journal, **vol. 895**(2), (2020), p. 83. URL <http://dx.doi.org/10.3847/1538-4357/ab89ad>.
- [70] **Riddhi Bandyopadhyay**, Yan Yang, William H. Matthaeus, Alexandros Chasapis, *et al.*, *In Situ Measurement of Curvature of Magnetic Field in Turbulent Space Plasmas: A Statistical Study*. The Astrophysical Journal Letters, **vol. 893**(1), (2020), p. L25. URL <http://dx.doi.org/10.3847/2041-8213/ab846e>.
- [71] William H. Matthaeus, Yan Yang, Minping Wan, Tulasi N. Parashar, *et al.*, *Pathways to Dissipation in Weakly Collisional Plasmas*. The Astrophysical Journal, **vol. 891**(1), (2020), p. 101. URL <http://dx.doi.org/10.3847/1538-4357/ab6d6a>.
- [72] T. N. Parashar, M. L. Goldstein, B. A. Maruca, W. H. Matthaeus, *et al.*, *Measures of Scale-dependent Alfvénicity in the First PSP Solar Encounter*. The Astrophysical Journal Supplement Series, **vol. 246**(2), (2020), p. 58. URL <http://dx.doi.org/10.3847/1538-4365/ab64e6>.
- [73] Rohit Chhiber, M L. Goldstein, B. A. Maruca, A. Chasapis, *et al.*, *Clustering of Intermittent Magnetic and Flow Structures near Parker Solar Probe’s First Perihelion—A Partial-variance-of-increments Analysis*. The Astrophysical Journal Supplement Series, **vol. 246**(2), (2020), p. 31. URL <http://dx.doi.org/10.3847/1538-4365/ab53d2>.
- [74] R. A. Qudsi, B. A. Maruca, W. H. Matthaeus, T. N. Parashar, *et al.*, *Observations of Heating along Intermittent Structures in the Inner Heliosphere from PSP Data*. The Astrophysical Journal Supplement Series, **vol. 246**(2), (2020), p. 46. URL <http://dx.doi.org/10.3847/1538-4365/ab5c19>.
- [75] **Riddhi Bandyopadhyay**, W. H. Matthaeus, T. N. Parashar, R. Chhiber, *et al.*, *Observations of Energetic-particle Population Enhancements along Intermittent Structures near the Sun from the Parker Solar Probe*. The Astrophysical Journal Supplement Series, **vol. 246**(2), (2020), p. 61. URL <http://dx.doi.org/10.3847/1538-4365/ab6220>.
- [76] **Riddhi Bandyopadhyay**, M. L. Goldstein, B. A. Maruca, W. H. Matthaeus, *et al.*, *Enhanced Energy Transfer Rate in Solar Wind Turbulence Observed near the Sun from Parker Solar Probe*. The Astrophysical Journal Supplement Series, **vol. 246**(2), (2020), p. 48. URL <http://dx.doi.org/10.3847/1538-4365/ab5dae>.
- [77] **Riddhi Bandyopadhyay**, William H. Matthaeus, Sean Oughton, and Minping Wan, *Evolution of Similarity Lengths in Anisotropic Magnetohydrodynamic Turbulence*. Journal of Fluid Mechanics, **vol. 876**, (2019), pp. 5. URL <http://dx.doi.org/10.1017/jfm.2019.513>.
- [78] **Riddhi Bandyopadhyay**, S. Oughton, M. Wan, W. H. Matthaeus, *et al.*, *Finite Dissipation in Anisotropic Magnetohydrodynamic Turbulence*. Physical Review X, **vol. 8**, (2018), p. 041052. URL <http://dx.doi.org/10.1103/PhysRevX.8.041052>.
- [79] Tulasi N. Parashar, Alexandros Chasapis, **R. Bandyopadhyay**, Rohit Chhiber, *et al.*, *Kinetic Range Spectral Features of Cross Helicity Using the Magnetospheric Multiscale Spacecraft*. Physical Review Letters, **vol. 121**, (2018), p. 265101. URL <http://dx.doi.org/10.1103/PhysRevLett.121.265101>.
- [80] R. Chhiber, A. Chasapis, **R. Bandyopadhyay**, T. N. Parashar, *et al.*, *Higher-Order Turbulence Statistics in the Earth’s Magnetosheath and the Solar Wind Using Magnetospheric Multiscale Observations*. Journal of Geophysical Research: Space Physics, **vol. 123**(12), (2018), pp. 9941. URL <http://dx.doi.org/10.1029/2018JA025768>.
- [81] B. A. Maruca, A. Chasapis, S. P. Gary, **R. Bandyopadhyay**, *et al.*, *MMS Observations of Beta-Dependent Constraints on Ion Temperature–Anisotropy in Earth’s Magnetosheath*. The Astrophysical Journal, **vol. 866**, (2018), p. 25. URL <http://dx.doi.org/10.3847/1538-4357/aaddfb>.
- [82] **Riddhi Bandyopadhyay**, A. Chasapis, R. Chhiber, T. N. Parashar, *et al.*, *Incompressive Energy Transfer in the Earth’s Magnetosheath: Magnetospheric Multiscale Observations*. The Astrophysical Journal, **vol. 866**(2), (2018), p. 106. URL <http://dx.doi.org/10.3847/1538-4357/aade04>.

- [83] **Riddhi Bandyopadhyay**, A. Chasapis, R. Chhiber, T. N. Parashar, *et al.*, *Solar Wind Turbulence Studies Using MMS Fast Plasma Investigation Data*. The Astrophysical Journal, **vol. 866**(2), (2018), p. 81. URL <http://dx.doi.org/doi.org/10.3847/1538-4357/aade93>.
- [84] **Riddhi Bandyopadhyay**, William H. Matthaeus, and Tulasi N. Parashar, *Single-Mode Nonlinear Langevin Emulation of Magnetohydrodynamic turbulence*. Phys. Rev. E, **vol. 97**, (2018), p. 053211. URL <http://dx.doi.org/10.1103/PhysRevE.97.053211>.
- [85] **Riddhi Bandyopadhyay** and Mahendra K. Verma, *Discrete Symmetries in Dynamo Reversals*. Physics of Plasmas, **vol. 24**(6), (2017), p. 062307. [1705.09630](https://doi.org/10.1063/1.4985307), URL <http://dx.doi.org/10.1063/1.4985307>.

UNREFEREED PUBLICATIONS

- [86] Sohom Roy, **Riddhi Bandyopadhyay**, William H Matthaeus, and Prayash Sharma Pyakurel, *Analysis of energy dissipation during transition from electron-only to ion-coupled reconnection*. AGU23.
- [87] Sohom Roy, **Riddhi Bandyopadhyay**, William H Matthaeus, and Yan Yang, *Investigating the scale-dependent conversion of turbulent energy in the magnetosheath*. AGU23.
- [88] **Riddhi Bandyopadhyay**, Sohom Roy, Yan Yang, Tulasi Parashar, *et al.*, *Turbulent Cascade and Proton–Electron Heating in Collisionless Plasmas*. AGU23.
- [89] Matthew R Argall, **Riddhi Bandyopadhyay**, Frederick D Wilder, and Stefan Eriksson, *Observations of young turbulence driven by reconnection at Earth’s magnetopause*. AGU23.
- [90] Bennett Maruca, Ramiz A Qudsi, Benjamin L Alterman, Brian Walsh, *et al.*, *The Trans-Heliospheric Survey: Trends in Plasma Parameters Across the Heliosphere*. AGU23.
- [91] Michael A Shay, Muhammad Bilal Khan, Colby C Haggerty, **Riddhi Bandyopadhyay**, *et al.*, *Statistics of Magnetic Reconnection in MHD Turbulence and its Implications for Dissipation*. AGU23.
- [92] Steven V Heuer, Kevin James Genestreti, James Burch, **Riddhi Bandyopadhyay**, *et al.*, *Plasma Heating and Energization During Guide Field Reconnection*. AGU23.
- [93] Francesco Pecora, Yan Yang, Alexandros Chasapis, Sergio Servidio, *et al.*, *Turbulence relaxation processes in the Earth’s magnetosheath*. AGU23.
- [94] Alexandros Chasapis, Tien Vo, Francesco Pecora, Sohom Roy, *et al.*, *Study of turbulence, dissipation, and particle energization in Earth’s magnetosheath with a dedicated survey by the Magnetospheric MultiScale mission*. AGU23.
- [95] Colby Haggerty, Riddhi Bandyopadhyay, Jason TenBarge, Paul Cassak, *et al.* (2023). *The Physics of Collisionless Dissipation in the Heliosphere*. In: Bulletin of the American Astronomical Society, vol. 55, p. 146. URL <http://dx.doi.org/10.3847/25c2cfeb.f79b0456>.
- [96] W. H. Matthaeus, S. Adhikari, **R. Bandyopadhyay**, M. R. Brown, *et al.* (2023). *The essential role of multi-point measurements in investigations of heliospheric turbulence, three-dimensional structure, and dynamics*. White Paper submitted to: Decadal Survey for Solar and Space Physics (Heliophysics) 2024-2033. arXiv admin note: substantial text overlap with arXiv:1903.06890, URL <http://dx.doi.org/10.48550/arXiv.2211.12676>.
- [97] Sohom Roy, **Riddhi Bandyopadhyay**, William Matthaeus, and Prayash Pyakurel, *Energy dissipation in electron-only reconnection*. Bulletin of the American Physical Society.
- [98] Juston Ward and **Riddhi Bandyopadhyay**, *A Search for Magnetic Switchbacks in Parker Solar Probe Data and a Statistical Analysis of Associated Solar Wind Tangential Velocity*. Bulletin of the American Physical Society.
- [99] James L Burch, Kevin James Genestreti, Steven Heuer, Alexandros Chasapis, *et al.* (2022). *Electron Heating in the Magnetic Reconnection Inflow Region*. In: Fall Meeting 2022. AGU.
- [100] Robert W Ebert, Frederic Allegrini, Fran Bagenal, **Riddhi Bandyopadhyay**, *et al.* (2022). *Observations of Jupiter’s Dusk Magnetopause at High Southern Latitude: Implications for 3-D Magnetosphere Structure and Solar Wind Interactions*. In: Fall Meeting 2022. AGU.

- [101] Sohom Roy, **Riddhi Bandyopadhyay**, William H Matthaeus, Alexandros Chasapis, *et al.* (2022). *Role of Turbulence and Reconnection in Dissipation in Plasmas*. In: Fall Meeting 2022. AGU.
- [102] William H Matthaeus, Rohit Chhiber, Arcadi V Usmanov, **Riddhi Bandyopadhyay**, *et al.* (2022). *A fragmented and extended Alfvén critical zone in the young solar wind*. In: Fall Meeting 2022. AGU.
- [103] Panisara Thepthong, Peera Pongkitiwanichakul, David J Ruffolo, Rungployphan Om Kieokaew, *et al.* (2022). *Scale-Dependent Alfvénicity of Super-Alfvénic and Sub-Alfvénic Solar Wind Observed during Parker Solar Probe’s 8th Orbit*. In: Fall Meeting 2022. AGU.
- [104] Luke Begley, **Riddhi Bandyopadhyay**, Bennett Maruca, David J McComas, *et al.* (2022). *Beta-dependent Constraints on Ion Temperature Anisotropy in Jupiter’s Magnetosheath*. In: Fall Meeting 2022. AGU.
- [105] Alexandros Chasapis, Luca Sorriso-Valvo, **Riddhi Bandyopadhyay**, Robert Ergun, *et al.* (2022). *Multi-point analysis of turbulence using colinear MMS observations in Earth’s magnetosheath*. In: Fall Meeting 2022. AGU.
- [106] Nawin Ngampoopun, David Ruffolo, **Riddhi Bandyopadhyay**, and William Matthaeus (2022). *Analysis of Turbulence Energy Transfer at an Interplanetary Shock Observed by MMS*. In: EGU General Assembly Conference Abstracts, pp. EGU22–6596.
- [107] **Riddhi Bandyopadhyay**, Sohom Roy, Tulasi Parashar, Subash Adhikari, *et al.*, *Energy transfer and proton-electron heating in turbulent plasmas*. The Third Triennial Earth-Sun Summit (TESS, vol. **54**(7)).
- [108] David Ruffolo, William Matthaeus, Panisara Thepthong, Peera Pongkitiwanichakul, *et al.*, *Alfvénicity of Velocity and Magnetic Field Increments Observed by Parker Solar Probe*. 44th COSPAR Scientific Assembly. Held 16-24 July, vol. **44**, (2022), p. 1418.
- [109] William H Matthaeus, **Riddhi Bandyopadhyay**, Yan Yang, Tulasi Parashar, *et al.*, *MMS Community Workshop*.
- [110] Manuel E Cuesta, Yan Yang, Francesco Pecora, William H Matthaeus, *et al.*, *Magnetic-field Line Curvature Observed by Magnetospheric Multiscale*.
- [111] Alexandros Chasapis, Robert E Ergun, Steven Schwartz, Narges Ahmadi, *et al.*, *MMS Science opportunities for a cross-scale colinear configuration campaign*.
- [112] **Riddhi Bandyopadhyay**, *Lessons from Sub-Alfvénic Solar Wind Observed by the Parker Solar Probe*. The Third Triennial Earth-Sun Summit (TESS, vol. **54**(7)).
- [113] Sohom Roy, **Riddhi Bandyopadhyay**, William H Matthaeus, Alexandros Chasapis, *et al.*, *Statistics of Energy Dissipation rate at reconnection sites*.
- [114] Rohit Chhiber, William Matthaeus, Arcadi Usmanov, Melvyn Goldstein, *et al.*, *Turbulence in the young solar wind-Parker Solar Probe observations and global modeling*. Bulletin of the American Physical Society.
- [115] Eric Zirnstein, Maher Dayeh, Rahul Kumar, Jacob Heerikhuisen, *et al.* (2021). *Proton Distribution in the Heliosheath: From Particle Acceleration to ENA Observation*. In: AGU Fall Meeting 2021. AGU.
- [116] Ramiz A Qudsi, **Riddhi Bandyopadhyay**, Bennett Maruca, and William H Matthaeus (2021). *Interplay Between Linear and Nonlinear Processes in Simulation and Space Plasmas*. In: AGU Fall Meeting 2021. AGU.
- [117] Yan Yang, William H Matthaeus, Tulasi N Parashar, **Riddhi Bandyopadhyay**, *et al.* (2021). *Cascade and Energy Balance in Plasma Turbulence: Scale Filtering and Structure Functions*. In: AGU Fall Meeting 2021. AGU.
- [118] Nawin Ngampoopun, David J Ruffolo, William H Matthaeus, **Riddhi Bandyopadhyay**, *et al.* (2021). *Analysis of Turbulence Energy Transfer at an Interplanetary Shock Observed by MMS*. In: AGU Fall Meeting 2021. AGU.
- [119] Nikos Sioulas, Marco CM Velli, William H Matthaeus, Loukas Vlahos, *et al.* (2021). *Statistical analysis of intermittent structures and their implications on heating during the first six PSP encounters*. In: AGU Fall Meeting 2021. AGU.
- [120] **Riddhi Bandyopadhyay**, David J McComas, Jamey R Szalay, Frederic Allegrini, *et al.* (2021). *Nature of Turbulence in the Jovian Magnetosheath from Juno Data*. In: AGU Fall Meeting 2021. AGU.
- [121] **Riddhi Bandyopadhyay**, Alexandros Chasapis, Yan Yang, Tulasi N Parashar, *et al.* (2021). *Energy Dissipation in Turbulent Plasmas*. In: AGU Fall Meeting 2021. AGU.

- [122] R. A. Qudsi, M. Richardson, H. DeWeese, J. A. Rueda, *et al.* (2020). *Magnetic Field topology reconstruction in a 3-D simulation box using Gaussian Process Regression*. In: AGU Fall Meeting Abstracts, vol. 2020, pp. NG004–0028. URL <https://ui.adsabs.harvard.edu/abs/2020AGUFMNG0040028Q>.
- [123] W. H. Matthaeus and **R. Bandyopadhyay** (2020). *Behavior of energetic particles in magnetic switchbacks observed by ISOIS in early PSP orbits*. In: AGU Fall Meeting Abstracts, vol. 2020, pp. SH049–0022. URL <https://ui.adsabs.harvard.edu/abs/2020AGUFMSH0490022M>.
- [124] M. E. Cuesta, T. Parashar, **R. Bandyopadhyay**, R. Chhiber, *et al.* (2020). *Magnetic Field Intermittency Statistics in the Expanding Solar Wind: Parker Solar Probe (Encounters 1-3), Helios 1 (0.3 to 1AU), and Voyager 1 (1 to 10AU)*. In: AGU Fall Meeting Abstracts, vol. 2020, pp. SH016–0013. URL <https://ui.adsabs.harvard.edu/abs/2020AGUFMSH0160013C>.
- [125] B. Maruca, R. A. Qudsi, R. Chhiber, **R. Bandyopadhyay**, *et al.* (2020). *Radial Trends in Plasma Parameters Across the Heliosphere*. In: AGU Fall Meeting Abstracts, vol. 2020, pp. SH017–0011. URL <https://ui.adsabs.harvard.edu/abs/2020AGUFMSH0170011M>.
- [126] Y. Wang, **R. Bandyopadhyay**, R. Chhiber, A. Chasapis, *et al.* (2020). *Statistical Survey of Collisionless Dissipation in the Terrestrial Magnetosheath*. In: AGU Fall Meeting Abstracts, vol. 2020, pp. SH016–0006. URL <https://ui.adsabs.harvard.edu/abs/2020AGUFMSH0160006W>.
- [127] A. V. Usmanov, R. Chhiber, W. H. Matthaeus, M. L. Goldstein, *et al.* (2020). *Global MHD Modeling of the Solar Corona and Solar Wind with Turbulent Transport and Heating: Comparisons with Parker Solar Probe Observations*. In: AGU Fall Meeting Abstracts, vol. 2020, pp. SH049–0011. URL <https://ui.adsabs.harvard.edu/abs/2020AGUFMSH0490011U>.
- [128] **R. Bandyopadhyay**, W. H. Matthaeus, T. N. Parashar, Y. Yang, *et al.* (2020). *Donald L. Turcotte Award: Pathways to Dissipation in Weakly Collisional Plasmas*. In: AGU Fall Meeting Abstracts, vol. 2020, pp. NG011–06. URL <https://ui.adsabs.harvard.edu/abs/2020AGUFMNG011..06B>.
- [129] P. Mostafavi, V.G. Merkin, E. Provornikova, N.E. Raouafi, *et al.*, *High-resolution Modeling of the Solar Wind Turbulence: from Global to Micro-scales*. White paper. URL <https://www.hou.usra.edu/meetings/helio2050/pdf/4069.pdf>.
- [130] M. Wan, C. Ying, Y. Yang, **R. Bandyopadhyay**, *et al.* (2019). *Von Karman Similarity Decay in Compressible Magnetohydrodynamic Turbulence*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH13E–3463. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH13E3463W>.
- [131] W. H. Matthaeus, **R. Bandyopadhyay**, T. Parashar, A. Chasapis, *et al.* (2019). *Statistics of Energetic Particles in the first Parker Solar Probe Orbit: Correlations and Association with Magnetic Structures*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH52A–07. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH52A..07M>.
- [132] **R. Bandyopadhyay**, M. L. Goldstein, B. Maruca, W. H. Matthaeus, *et al.* (2019). *MHD-Scale Energy Transfer in the Inner Heliosphere from PSP observations*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH21C–3318. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH21C3318B>.
- [133] R. A. Qudsi, B. Maruca, W. H. Matthaeus, T. Parashar, *et al.* (2019). *Intermittent heating in the inner Heliosphere: PSP observations*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH21C–3317. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH21C3317Q>.
- [134] A. Chasapis, **R. Bandyopadhyay**, W. H. Matthaeus, R. Chhiber, *et al.* (2019). *Multi-point observations of solar wind turbulence during the MMS Solar wind turbulence campaign*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH11B–07. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH11B..07C>.
- [135] **R. Bandyopadhyay**, Y. Yang, M. Wan, W. H. Matthaeus, *et al.* (2019). *Curvature of Magnetic Field in Plasma Turbulence*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH11B–01. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH11B..01B>.
- [136] R. Chhiber, M. L. Goldstein, W. H. Matthaeus, **R. Bandyopadhyay**, *et al.* (2019). *Waiting time (distance) distributions of magnetic field and velocity PVI events during the first Parker Solar Probe encounter*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH13C–3451. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH13C3451C>.

- [137] W. H. Matthaeus, D. J. Ruffolo, C. DeForest, T. Parashar, *et al.* (2019). *Flocculation, switchbacks, and loss of Alfvénicity: Indicators of shear-driven turbulence in the young solar wind?* In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH53B–3374. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH53B3374M>.
- [138] B. Maruca, R. A. Qudsi, **R. Bandyopadhyay**, T. Parashar, *et al.* (2019). *A Statistical Study of Proton Microinstabilities and Nonlinear Effects in Space Plasmas*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH51D–3310. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH51D3310M>.
- [139] A. Chasapis, R. Chhiber, **R. Bandyopadhyay**, D. Malaspina, *et al.* (2019). *On the validity of the Taylor Hypothesis in the inner heliosphere as observed by the Parker solar probe*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH21C–3320. URL <https://ui.adsabs.harvard.edu/abs/2019AGUFMSH21C3320C>.
- [140] **R Bandyopadhyay**, TN Parashar, ML Goldstein, B Maruca, *et al.* (2019). *Alfvénicity in PSP observations: comparing different measures*. In: AGU Fall Meeting Abstracts, vol. 2019, pp. SH13C–3436.
- [141] W. H. Matthaeus, **R. Bandyopadhyay**, M. R. Brown, J. Borovsky, *et al.*, [Plasma 2020 Decadal] *The essential role of multi-point measurements in turbulence investigations: the solar wind beyond single scale and beyond the Taylor Hypothesis*. arXiv e-prints, arXiv:1903.06890. 1903.06890, URL <https://ui.adsabs.harvard.edu/abs/2019arXiv190306890M>.
- [142] **Riddhi Bandyopadhyay**, R. Qudsi, W. H. Matthaeus, T. N. Parashar, *et al.* (2019). *Dominance of Local Turbulence over Ion-Driven Micro-Instabilities in Turbulent Plasmas*. In: Solar Heliospheric and Interplanetary Environment (SHINE 2019), p. 15. URL <https://ui.adsabs.harvard.edu/abs/2019shin.confE..15B>.
- [143] Mark Pultrone, B. A. Maruca, R. A. Qudsi, A. Chasapis, *et al.* (2019). *MMS Observations of Ion and Electron Temperature Anisotropy in the Magnetosheath*. In: Solar Heliospheric and Interplanetary Environment (SHINE 2019), p. 171. URL <https://ui.adsabs.harvard.edu/abs/2019shin.confE.171P>.
- [144] M. A. Shay, **R. Bandyopadhyay**, T. Parashar, W. H. Matthaeus, *et al.* (2018). *Statistics of Magnetic Reconnection in Turbulence and its Effect on Plasma Heating*. In: AGU Fall Meeting Abstracts, vol. 2018, pp. SH31B–3621. URL <https://ui.adsabs.harvard.edu/abs/2018AGUFMSH31B3621S>.
- [145] T. N. Parashar, A. Chasapis, **R. Bandyopadhyay**, R. Chhiber, *et al.* (2018). *Cross helicity in the kinetic range of plasma turbulence using MMS*. In: AGU Fall Meeting Abstracts, vol. 2018, pp. SH21C–3295. URL <https://ui.adsabs.harvard.edu/abs/2018AGUFMSH21C3295P>.
- [146] J. Fanelle, M. A. Shay, W. H. Matthaeus, A. Chasapis, *et al.* (2018). *Estimating Current Density with MMS Data in Near-Earth Space*. In: AGU Fall Meeting Abstracts, vol. 2018, pp. SM31B–07. URL <https://ui.adsabs.harvard.edu/abs/2018AGUFMSM31B..07F>.
- [147] **R. Bandyopadhyay**, W. H. Matthaeus, A. Chasapis, R. Chhiber, *et al.* (2018). *Turbulence Studies using MMS: data issues and physics*. In: AGU Fall Meeting Abstracts, vol. 2018, pp. SH21C–3294. URL <https://ui.adsabs.harvard.edu/abs/2018AGUFMSH21C3294B>.
- [148] **R. Bandyopadhyay**, A. Chasapis, R. Chhiber, T. N. Parashar, *et al.* (2018). *Incompressible Energy Transfer in the Earth’s Magnetosheath: MMS Observations*. In: Solar Heliospheric and Interplanetary Environment (SHINE 2018), p. 27. URL <https://ui.adsabs.harvard.edu/abs/2018shin.confE..27B>.
- [149] **Riddhi Bandyopadhyay** (2017). *Modeling turbulence using Langevin equation*. In: Solar Heliospheric and Interplanetary Environment (SHINE 2017), p. 153. URL <https://ui.adsabs.harvard.edu/abs/2017shin.confE.153B>.
- [150] T. N. Parashar, **R. Bandyopadhyay**, and W. H. Matthaeus (2017). *Modeling the Behaviour of Inertial Modes in MHD Turbulence*. In: AGU Fall Meeting Abstracts, vol. 2017, pp. SH11A–2428. URL <https://ui.adsabs.harvard.edu/abs/2017AGUFMSH11A2428P>.

TEACHING EXPERIENCE

Instructor (University of Delaware)

PHYS635 Space Physics (Graduate level course on physical processes in our solar system)

Spring 2026

Guest Lecturer (University of Delaware)

Plasma Physics I (Graduate level course on fluids and magnetohydrodynamics)	Fall 2019
PHYS638 Turbulent Flows (Graduate level course on fundamental characteristics of turbulence)	Spring 2025
PHYS809 Electrodynamics I (Graduate level course on Electrodynamics)	Fall 2025

Teaching Assistant (University of Delaware)

Introductory Physics I (Introduction to classical physics and fluid mechanics for students in the life and environmental sciences)	2017 Winter, 2017 Fall, 2018 Winter
Introductory Physics II (Introduction to electromagnetism and optics for students in the life and environmental sciences)	2017 Spring
Fundamentals of Physics II (Introduction to electromagnetism for students in the physical sciences and engineering)	2017 Summer, 2017 Fall

SYNERGISTIC ACTIVITIES

Reviewer for The Astrophysical Journal (ApJ), The Astrophysical Journal Letters (ApJL), Journal of Geophysical Research: Space Physics (JGR), Geophysical Research Letters (GRL), Monthly Notices of the Royal Astronomical Society (MNRAS), Physics of Plasmas (PoP), Physical Review Letters (PRL), Earth and Space Science (ESS), Physical Review E (PRE), Scientific Reports, Astronomy & Astrophysics (A & A), Nature Communications, Journal of Plasma Physics (JPP), Space Science Reviews (SSR) (~ 10 per year).

Review Panels

NASA Review Panelist	2021, 2022, 2023
NSF Review Panelist	2025
National Research and Development Agency, Government of Chile	2022, 2024, 2025
NASA External Reviewer	2020, 2024, 2026
NASA Executive Secretary	2020

Conference Panel Organization

Magnetospheric Multiscale (MMS) Mission Science Working Team meeting	Spring 2020, Fall 2020, Fall 2024
Interstellar Boundary Explorer (IBEX) Mission Science Team meeting	2021, 2022
Solar Heliospheric and INTERplanetary Environment (SHINE) workshop	2022, 2023, 2024

Graduate Student & Dissertation Advising**Next Position**

William Taranto (Ph.D. expected 2030, University of Delaware)	N/A (ongoing)
Sohom Roy (Ph.D. December 2024, University of Delaware)	postdoc at Space Research Institute (Institut für Weltraumforschung, IWF), Graz, Austria
Camryn Phillips (Fall 2021 and Spring 2022 semester projects, Princeton University)	changed field

Undergraduate Student Advising**Next Position**

Yanwen Wang, '22 (2019 Sophomore Project), University of Delaware	Ph.D. student at the University of Maryland, College Park
Luke Begley, '22 (2020 Junior Project), Princeton University	joined industry
Siegfried Gawenda, '23 (2022 Spring Junior Project), Princeton University	Ph.D. student at the University of Texas Rio Grande Valley
Ryan LoRusso, '24 (2022 Fall Junior Project), Princeton University	Ph.D. student at Indiana University at Bloomington
Cole Meyer, '24 (2022 Fall Junior Project), Princeton University	Ph.D. student at Lunar and Planetary Laboratory at the University of Arizona

Juston Ward, '24 (Princeton University Undergraduate Summer Research Program via National Radio Astronomy Observatory/National Astronomy Consortium; 2023&2024 Summer), Sam Houston State University Ph.D. student at Baylor University
Alex Bruce, '26 (2025 Winter Project), University of Delaware –

Thesis Committee

External member for dissertation proposal of Zac Bailey at the Institute for Astronomy, University of Hawai'i (2026)

Research Readiness Exam Committee Chair for Zihang Cheng (advisor: Prof. Yan Yang), University of Delaware (2026)

Public Outreach

Lecture and Interaction at the "[Science Under the Stars](#)" series organized by the New Jersey State Museum.

Miscellaneous

Invited member of the Solar Heliospheric and Interplanetary Environment (SHINE) workshop Steering Committee.

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